# Assignment 2 Report: Simple Search Engine using Hadoop MapReduce

Assignment link: https://firas-jolha.github.io/bigdata/html/bs/BS%20-%20Assignment%202%20-%20Simple%20Search%20Engine%20using%20Hadoop%20MapReduce.html

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## Methodology

In this assignment, I built a small search engine by using Hadoop MapReduce, Cassandra, and PySpark together. The main goal was to take a Parquet dataset, turn it into text documents, build an inverted index, store the final index in Cassandra, and rank query results with BM25.

The dataset file used in my project is `a.parquet`, and I place it in `/app/a.parquet`. The main script is `app.sh`. When the project starts, this script starts Hadoop and YARN, creates the Python virtual environment, installs the needed packages, runs data preparation, builds the index, stores it in Cassandra, and runs two example queries.

For data preparation, I used `prepare_data.sh` and `prepare_data.py`. Spark reads the Parquet file and keeps the columns `id`, `title`, and `text`. After that, rows with missing values or empty text are removed. From the valid rows, the program keeps 1000 documents. Each document is written as a UTF-8 text file with the format `<doc_id>_<doc_title>.txt`. These files are uploaded to HDFS under `/data`.

After creating the text files, the same documents are converted into the format required for indexing. Each line becomes `<doc_id><tab><doc_title><tab><doc_text>`, and this output is written to HDFS under `/input/data` as one partition.

The indexing part is done by `create_index.sh` and the Hadoop Streaming scripts in `app/mapreduce`. In the first MapReduce pipeline, `mapper1.py` reads each document, tokenizes the text, and counts term frequency inside the document. Then `reducer1.py` groups records by term and creates the main posting data and vocabulary data. In the second pipeline, `mapper2.py` and `reducer2.py` calculate corpus statistics such as total number of documents and average document length. These values are needed later by BM25.

After the Hadoop jobs finish, the outputs are split into four final parts:

- vocabulary
- index
- documents
- stats

These files are stored in HDFS under `/indexer/vocabulary`, `/indexer/index`, `/indexer/documents`, and `/indexer/stats`.

For storage, I used `store_index.sh` and `app.py`. The program connects to the Cassandra container and creates the keyspace `search_engine`. Inside this keyspace, the project stores four tables: `vocabulary`, `postings`, `documents`, and `corpus_stats`. The vocabulary table stores document frequency for each term.

The postings table stores term frequency and document information. The documents table stores document title and length. The corpus stats table stores values like `total_docs` and `avg_doc_length`.

For search, I used `search.sh` and `query.py`. The query text is passed to `search.sh` as a user argument, and the script forwards it to `query.py` through stdin. Then `query.py` tokenizes the text with the same regular expression used during indexing. After that, the program reads the needed rows from Cassandra and calculates BM25 scores with PySpark. The BM25 formula uses term frequency, document frequency, document length, and average document length. After scoring, the top 10 results are printed in the format `doc_id<TAB>title`.

The search script runs on YARN in cluster mode. This was useful for me because it showed how the search step can also run as a distributed job, not only the indexing step.

For the optional task, I added `add_to_index.sh`. This script takes one local text file, uploads it to HDFS `/data`, rebuilds `/input/data`, reruns the indexing steps, and updates the Cassandra tables. The script uses the local file name as the document title and creates a new document id from the current time. Because of that, the search engine can include a new file after the first index has already been created.

## Demonstration

To run the repository, use:

```
git lfs pull
docker compose up
```

We need `git lfs pull` first because the large `a.parquet` file is stored with Git LFS.

The automatic flow is:

1. start Hadoop and YARN
2. create `.venv`
3. install packages from `requirements.txt`
4. run `prepare_data.sh`
5. run `index.sh`
6. run `search.sh "computer science"`
7. run `search.sh "history"`

For manual checking after startup, I used these commands:

```
docker exec -it cluster-master bash
cd /app
hdfs dfs -ls /data
hdfs dfs -ls /input/data
hdfs dfs -ls /indexer
hdfs dfs -text /indexer/stats/part-*
bash search.sh "computer science"
bash search.sh "history"
```

These commands are also part of the normal automatic flow in `app.sh`. I ran them manually for the screenshots only because it was easier to capture the output step by step without the extra service startup messages.

The `search.sh` script takes the query as an argument, then sends it to `query.py`.

In the completed run, the project created `/data`, `/input/data`, and `/indexer` in HDFS, and Cassandra stored the final index tables. The corpus statistics showed `total_docs = 1000` and `avg_doc_length = 575.759`, so the indexing step finished correctly.

For testing the search engine, I used the queries `computer science` and `history`. The query `history` gave more clearly related results. Some of the returned titles were:

- `A Brief History of Time`
- `A Briefer History of Time`
- `A Brief History of Everyone Who Ever Lived`

I looked at the returned documents to understand why they were ranked. For `history`, the results were easier to understand because the word `history` appears directly in both the titles and the document text. For example, `A Brief History of Chinese Fiction` is a book about the history of Chinese fiction, `A Brief History of Everyone Who Ever Lived` explicitly says it is about human genetics and human history, and `A Chance to Make History` also repeats the same word in the title and body. Because of that, BM25 gives these documents strong scores for the query term.

The query `computer science` was more mixed, but I could still see why some of the returned entities appeared. The top result, `A B M Shawkat Ali`, is a good example because the article text says that he is a `computer scientist`, that he studied `computer science and engineering`, and that he works in `data science`. Another result, `A Can of Paint`, looks strange at first, but the text clearly calls it a `science fiction` short story, so it can still get a score from the word `science`. `A Brief History of Time (film)` and `A Briefer History of Time` also contain science-related wording in their texts. This showed me that my search engine is ranking by the full article text stored in the index, not just by the title, and it also scores the query terms separately instead of requiring the exact phrase `computer science`.

For the optional task, a new text file can be added with:

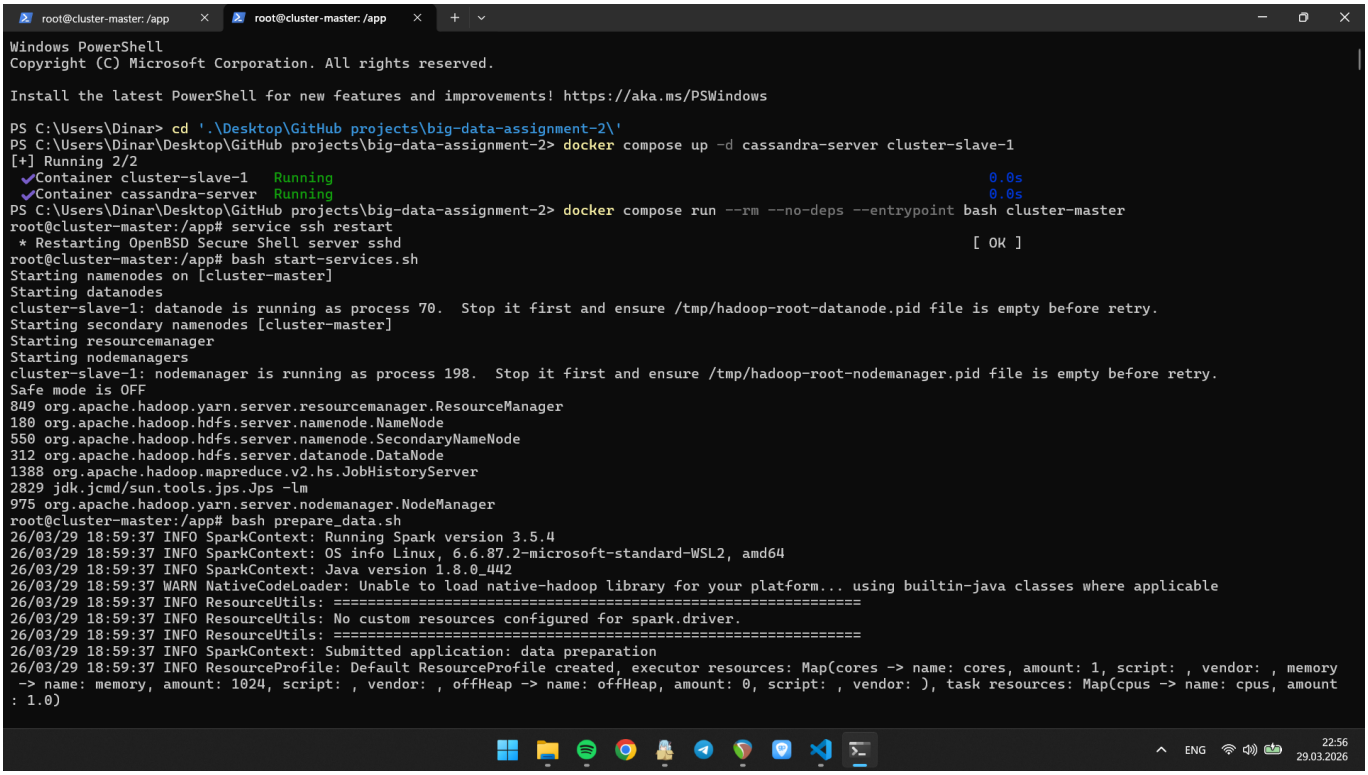
```
docker exec -it cluster-master bash
cd /app
printf 'This is a new document for the optional task.\n' > /app/New_Document.txt
bash add_to_index.sh /app/New_Document.txt
bash search.sh "new document"
```

This command adds the file into the existing pipeline, rebuilds the input for indexing, and updates the Cassandra data. In my script, the file name becomes the document title, and a new document id is generated automatically.

I also tested the optional task with a file called `New_Document.txt`. After running `add_to_index.sh`, the Cassandra statistics changed from `total_docs = 1000` to `total_docs = 1001`, so the new file was included in the index.

The following screenshots are from my successful runs:

I also saved the full terminal output from the successful manual run in `screenshots/logs.txt`, so the complete indexing and query logs are available in text form too.

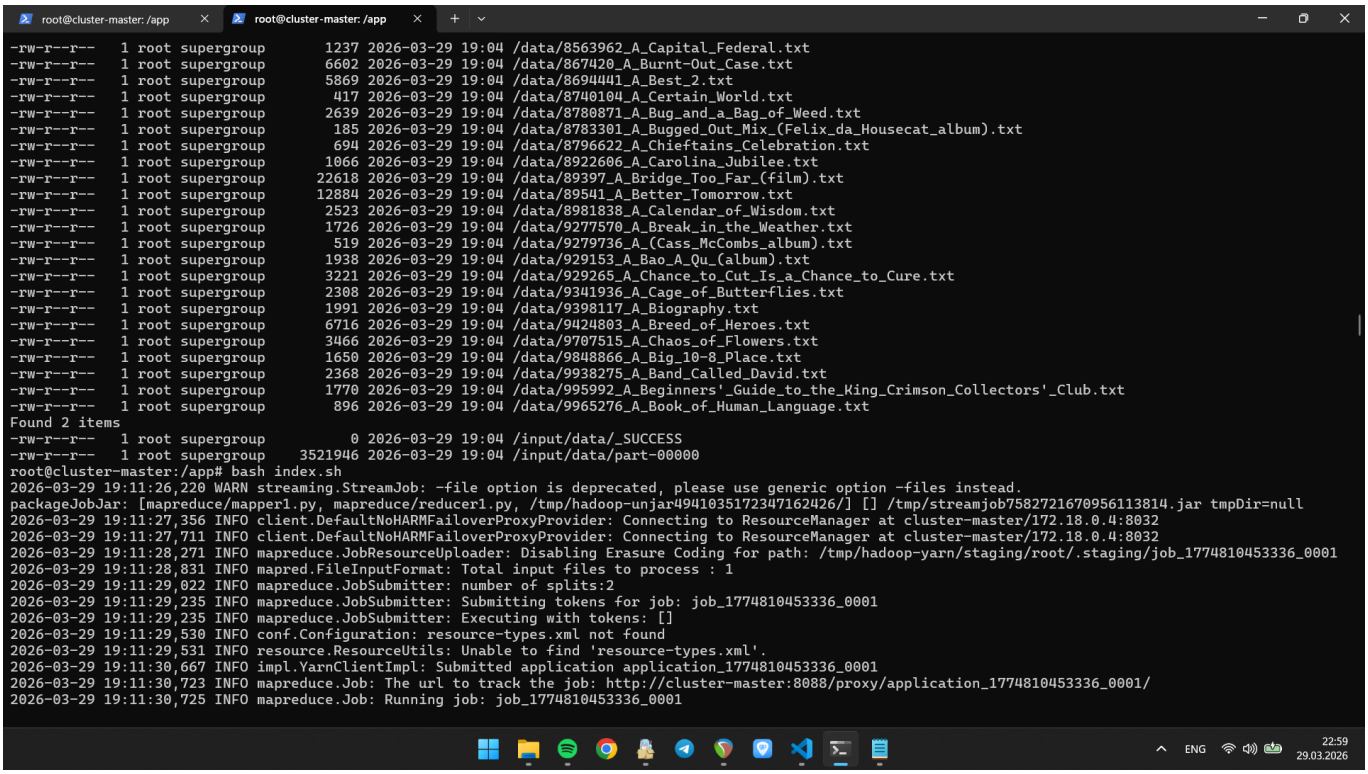


```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Dinar> cd '.\Desktop\GitHub projects\big-data-assignment-2\'
PS C:\Users\Dinar\Desktop\GitHub projects\big-data-assignment-2> docker compose up -d cassandra-server cluster-slave-1
[+] Running 2/2
 ✔Container cluster-slave-1   Running                                                                    0.0s
 ✔Container cassandra-server  Running                                                                    0.0s
PS C:\Users\Dinar\Desktop\GitHub projects\big-data-assignment-2> docker compose run --rm --no-deps --entrypoint bash cluster-master
root@cluster-master:/app# service ssh restart
 * Restarting OpenBSD Secure Shell server sshd                                                          [ OK ]
root@cluster-master:/app# bash start-services.sh
Starting namenodes on [cluster-master]
Starting datanodes
cluster-slave-1: datanode is running as process 70.  Stop it first and ensure /tmp/hadoop-root-datanode.pid file is empty before retry.
Starting secondary namenodes [cluster-master]
Starting resourcemanager
Starting nodemanagers
cluster-slave-1: nodemanager is running as process 198.  Stop it first and ensure /tmp/hadoop-root-nodemanager.pid file is empty before retry.
Safe mode is OFF
849 org.apache.hadoop.yarn.server.resourcemanager.ResourceManager
180 org.apache.hadoop.hdfs.server.namenode.NameNode
550 org.apache.hadoop.hdfs.server.namenode.SecondaryNameNode
312 org.apache.hadoop.hdfs.server.datanode.DataNode
1388 org.apache.hadoop.mapreduce.v2.hs.JobHistoryServer
2829 jdk.jcmd/sun.tools.jps.Jps -lm
975 org.apache.hadoop.yarn.server.nodemanager.NodeManager
root@cluster-master:/app# bash prepare_data.sh
26/03/29 18:59:37 INFO SparkContext: Running Spark version 3.5.4
26/03/29 18:59:37 INFO SparkContext: OS info Linux, 6.6.87.2-microsoft-standard-WSL2, amd64
26/03/29 18:59:37 INFO SparkContext: Java version 1.8.0_442
26/03/29 18:59:37 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
26/03/29 18:59:37 INFO ResourceUtils: ==============================================================
26/03/29 18:59:37 INFO ResourceUtils: No custom resources configured for spark.driver.
26/03/29 18:59:37 INFO ResourceUtils: ==============================================================
26/03/29 18:59:37 INFO SparkContext: Submitted application: data preparation
26/03/29 18:59:37 INFO ResourceProfile: Default ResourceProfile created, executor resources: Map(cores -> name: cores, amount: 1, script: , vendor: , memory
 -> name: memory, amount: 1024, script: , vendor: , offHeap -> name: offHeap, amount: 0, script: , vendor: ), task resources: Map(cpus -> name: cpus, amount
: 1.0)
```

This screenshot shows the manual run after entering `cluster-master`, starting Hadoop services, and launching data preparation.



```
-rw-r--r--   1 root supergroup       1237 2026-03-29 19:04 /data/8563962_A_Capital_Federal.txt
-rw-r--r--   1 root supergroup       6602 2026-03-29 19:04 /data/867420_A_Burnt-Out_Case.txt
-rw-r--r--   1 root supergroup       5869 2026-03-29 19:04 /data/8694441_A_Best_2.txt
-rw-r--r--   1 root supergroup        417 2026-03-29 19:04 /data/8740104_A_Certain_World.txt
-rw-r--r--   1 root supergroup       2639 2026-03-29 19:04 /data/8780871_A_Bug_and_a_Bag_of_Weed.txt
-rw-r--r--   1 root supergroup        185 2026-03-29 19:04 /data/8783301_A_Bugged_Out_Mix_(Felix_da_Housecat_album).txt
-rw-r--r--   1 root supergroup        694 2026-03-29 19:04 /data/8796622_A_Chieftains_Celebration.txt
-rw-r--r--   1 root supergroup       1066 2026-03-29 19:04 /data/8922606_A_Carolina_Jubilee.txt
-rw-r--r--   1 root supergroup      22618 2026-03-29 19:04 /data/89397_A_Bridge_Too_Far_(film).txt
-rw-r--r--   1 root supergroup      12884 2026-03-29 19:04 /data/89541_A_Better_Tomorrow.txt
-rw-r--r--   1 root supergroup       2523 2026-03-29 19:04 /data/8981838_A_Calendar_of_Wisdom.txt
-rw-r--r--   1 root supergroup       1726 2026-03-29 19:04 /data/9277570_A_Break_in_the_Weather.txt
-rw-r--r--   1 root supergroup        519 2026-03-29 19:04 /data/9279736_A_(Cass_McCombs_album).txt
-rw-r--r--   1 root supergroup       1938 2026-03-29 19:04 /data/929153_A_Bao_A_Qu_(album).txt
-rw-r--r--   1 root supergroup       3221 2026-03-29 19:04 /data/929265_A_Chance_to_Cut_Is_a_Chance_to_Cure.txt
-rw-r--r--   1 root supergroup       2308 2026-03-29 19:04 /data/9341936_A_Cage_of_Butterflies.txt
-rw-r--r--   1 root supergroup       1991 2026-03-29 19:04 /data/9398117_A_Biography.txt
-rw-r--r--   1 root supergroup       6716 2026-03-29 19:04 /data/9424803_A_Breed_of_Heroes.txt
-rw-r--r--   1 root supergroup       3466 2026-03-29 19:04 /data/9707515_A_Chaos_of_Flowers.txt
-rw-r--r--   1 root supergroup       1650 2026-03-29 19:04 /data/9848866_A_Big_10-8_Place.txt
-rw-r--r--   1 root supergroup       2368 2026-03-29 19:04 /data/9938275_A_Band_Called_David.txt
-rw-r--r--   1 root supergroup       1770 2026-03-29 19:04 /data/995992_A_Beginners'_Guide_to_the_King_Crimson_Collectors'_Club.txt
-rw-r--r--   1 root supergroup        896 2026-03-29 19:04 /data/9965276_A_Book_of_Human_Language.txt
Found 2 items
-rw-r--r--   1 root supergroup          0 2026-03-29 19:04 /input/data/_SUCCESS
-rw-r--r--   1 root supergroup    3521946 2026-03-29 19:04 /input/data/part-00000
root@cluster-master:/app# bash index.sh
2026-03-29 19:11:26,220 WARN streaming.StreamJob: -file option is deprecated, please use generic option -files instead.
packageJobJar: [mapreduce/mapper1.py, mapreduce/reducer1.py, /tmp/hadoop-unjar4941035172347162426/] [] /tmp/streamjob7582721670956113814.jar tmpDir=null
2026-03-29 19:11:27,356 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at cluster-master/172.18.0.4:8032
2026-03-29 19:11:27,711 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at cluster-master/172.18.0.4:8032
2026-03-29 19:11:28,271 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/root/.staging/job_1774810453336_0001
2026-03-29 19:11:28,831 INFO mapred.FileInputFormat: Total input files to process : 1
2026-03-29 19:11:29,022 INFO mapreduce.JobSubmitter: number of splits:2
2026-03-29 19:11:29,235 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1774810453336_0001
2026-03-29 19:11:29,235 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-03-29 19:11:29,530 INFO conf.Configuration: resource-types.xml not found
2026-03-29 19:11:29,531 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-03-29 19:11:30,667 INFO impl.YarnClientImpl: Submitted application application_1774810453336_0001
2026-03-29 19:11:30,723 INFO mapreduce.Job: The url to track the job: http://cluster-master:8088/proxy/application_1774810453336_0001/
2026-03-29 19:11:30,725 INFO mapreduce.Job: Running job: job_1774810453336_0001
```

This screenshot shows the beginning of `bash index.sh` and the first Hadoop Streaming pipeline starting.

```
root@cluster-master: /app    root@cluster-master: /app
                Input split bytes=222
                Combine input records=0
                Combine output records=0
                Reduce input groups=1
                Reduce shuffle bytes=13046
                Reduce input records=1000
                Reduce output records=1
                Spilled Records=2000
                Shuffled Maps =2
                Failed Shuffles=0
                Merged Map outputs=2
                GC time elapsed (ms)=178
                CPU time spent (ms)=4110
                Physical memory (bytes) snapshot=933376000
                Virtual memory (bytes) snapshot=7773687808
                Total committed heap usage (bytes)=770703360
                Peak Map Physical memory (bytes)=345317376
                Peak Map Virtual memory (bytes)=2590130176
                Peak Reduce Physical memory (bytes)=245690368
                Peak Reduce Virtual memory (bytes)=2593955840
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=16898728
        File Output Format Counters
                Bytes Written=20
2026-03-29 19:12:19,893 INFO streaming.StreamJob: Output directory: /tmp/indexer/pipeline2
Found 4 items
drwxr-xr-x   - root supergroup          0 2026-03-29 19:12 /indexer/documents
drwxr-xr-x   - root supergroup          0 2026-03-29 19:12 /indexer/index
drwxr-xr-x   - root supergroup          0 2026-03-29 19:12 /indexer/stats
drwxr-xr-x   - root supergroup          0 2026-03-29 19:12 /indexer/vocabulary
done storing index
root@cluster-master:/app# bash search.sh "computer science"
26/03/29 19:24:38 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
26/03/29 19:24:39 INFO DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at cluster-master/172.18.0.4:8032
```

This screenshot shows the end of the indexing step with the `/indexer` folders and the final `done storing index` message.

```
root@cluster-master: /app    root@cluster-master: /app
done storing index
root@cluster-master:/app# bash search.sh "computer science"
26/03/29 19:24:38 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
26/03/29 19:24:39 INFO DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at cluster-master/172.18.0.4:8032
26/03/29 19:24:40 INFO Configuration: resource-types.xml not found
26/03/29 19:24:40 INFO ResourceUtils: Unable to find 'resource-types.xml'.
26/03/29 19:24:40 INFO Client: Verifying our application has not requested more than the maximum memory capability of the cluster (8192 MB per container)
26/03/29 19:24:40 INFO Client: Will allocate AM container, with 896 MB memory including 384 MB overhead
26/03/29 19:24:40 INFO Client: Setting up container launch context for our AM
26/03/29 19:24:40 INFO Client: Setting up the launch environment for our AM container
26/03/29 19:24:40 INFO Client: Preparing resources for our AM container
26/03/29 19:24:40 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/HikariCP-2.5.1.jar
26/03/29 19:24:40 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/JLargeArrays-1.5.jar
26/03/29 19:24:40 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/JTransforms-3.1.jar
26/03/29 19:24:40 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/RoaringBitmap-0.9.45
.jar
26/03/29 19:24:40 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/ST4-4.0.4.jar
26/03/29 19:24:40 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/activation-1.1.1.jar
26/03/29 19:24:40 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/aircompressor-0.27.j
ar
26/03/29 19:24:40 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/algebra_2.12-2.0.1.j
ar
26/03/29 19:24:40 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/annotations-17.0.0.j
ar
26/03/29 19:24:40 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/antlr-runtime-3.5.2.
jar
26/03/29 19:24:40 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/antlr4-runtime-4.9.3
.jar
26/03/29 19:24:40 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/aopalliance-repackag
ed-2.6.1.jar
26/03/29 19:24:40 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/arpack-3.0.3.jar
26/03/29 19:24:40 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/arpack_combined_all-
0.1.jar
26/03/29 19:24:40 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/arrow-format-12.0.1.
jar
26/03/29 19:24:40 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/arrow-memory-core-12
.0.1.jar
26/03/29 19:24:40 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/arrow-memory-netty-1
2.0.1.jar
26/03/29 19:24:40 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/arrow-vector-12.0.1.
jar
```

This screenshot shows the first query run for `computer science` while the Spark job is being launched on YARN.

```
root@cluster-master: /app    root@cluster-master: /app
         queue: default
         start time: 1774812281669
         final status: UNDEFINED
         tracking URL: http://cluster-master:8088/proxy/application_1774810453336_0003/
         user: root
26/03/29 19:24:53 INFO Client: Application report for application_1774810453336_0003 (state: RUNNING)
26/03/29 19:24:53 INFO Client:
         client token: N/A
         diagnostics: N/A
         ApplicationMaster host: cluster-slave-1
         ApplicationMaster RPC port: 38627
         queue: default
         start time: 1774812281669
         final status: UNDEFINED
         tracking URL: http://cluster-master:8088/proxy/application_1774810453336_0003/
         user: root
26/03/29 19:25:16 INFO Client: Application report for application_1774810453336_0003 (state: FINISHED)
26/03/29 19:25:16 INFO Client:
         client token: N/A
         diagnostics: N/A
         ApplicationMaster host: cluster-slave-1
         ApplicationMaster RPC port: 38627
         queue: default
         start time: 1774812281669
         final status: SUCCEEDED
         tracking URL: http://cluster-master:8088/proxy/application_1774810453336_0003/
         user: root
26/03/29 19:25:16 INFO ShutdownHookManager: Shutdown hook called
26/03/29 19:25:16 INFO ShutdownHookManager: Deleting directory /tmp/spark-d3e8cf7c-e746-470d-90f5-d9603bc1346c
26/03/29 19:25:16 INFO ShutdownHookManager: Deleting directory /tmp/spark-203ce074-10ac-4d6b-aeee-a060782df3e7
48227028        A B M Shawkat Ali
51864384        A Can of Paint
1443000 A Brief History of Time (film)
39271639        A Boy and His Atom
43789955        A Braver Thing
15812271        A Cabin on the Coast
58539404        A Birthday
5259052 A Briefer History of Time (Schulman book)
651789  A Calculus of Angels
54088753        A Beautiful Star
root@cluster-master:/app# bash search.sh "history"
```

This screenshot shows the final ranked results for the query `computer science`.

```
root@cluster-master: /app    root@cluster-master: /app
54088753        A Beautiful Star
root@cluster-master:/app# bash search.sh "history"
26/03/29 19:40:21 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
26/03/29 19:40:22 INFO DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at cluster-master/172.18.0.4:8032
26/03/29 19:40:23 INFO Configuration: resource-types.xml not found
26/03/29 19:40:23 INFO ResourceUtils: Unable to find 'resource-types.xml'.
26/03/29 19:40:23 INFO Client: Verifying our application has not requested more than the maximum memory capability of the cluster (8192 MB per container)
26/03/29 19:40:23 INFO Client: Will allocate AM container, with 896 MB memory including 384 MB overhead
26/03/29 19:40:23 INFO Client: Setting up container launch context for our AM
26/03/29 19:40:23 INFO Client: Setting up the launch environment for our AM container
26/03/29 19:40:23 INFO Client: Preparing resources for our AM container
26/03/29 19:40:23 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/HikariCP-2.5.1.jar
26/03/29 19:40:23 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/JLargeArrays-1.5.jar
26/03/29 19:40:23 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/JTransforms-3.1.jar
26/03/29 19:40:23 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/RoaringBitmap-0.9.45
.jar
26/03/29 19:40:23 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/ST4-4.0.4.jar
26/03/29 19:40:23 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/activation-1.1.1.jar
26/03/29 19:40:23 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/aircompressor-0.27.j
ar
26/03/29 19:40:23 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/algebra_2.12-2.0.1.j
ar
26/03/29 19:40:23 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/annotations-17.0.0.j
ar
26/03/29 19:40:23 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/antlr-runtime-3.5.2.
jar
26/03/29 19:40:23 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/antlr4-runtime-4.9.3
.jar
26/03/29 19:40:23 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/aopalliance-repackag
ed-2.6.1.jar
26/03/29 19:40:23 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/arpack-3.0.3.jar
26/03/29 19:40:23 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/arpack_combined_all-
0.1.jar
26/03/29 19:40:23 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/arrow-format-12.0.1.
jar
26/03/29 19:40:23 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/arrow-memory-core-12
.0.1.jar
26/03/29 19:40:23 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/arrow-memory-netty-1
2.0.1.jar
26/03/29 19:40:23 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/arrow-vector-12.0.1.
jar
```

This screenshot shows the second query run for `history` during the YARN job startup.

```
root@cluster-master: /app    root@cluster-master: /app
         ApplicationMaster RPC port: -1
         queue: default
         start time: 1774813224861
         final status: UNDEFINED
         tracking URL: http://cluster-master:8088/proxy/application_1774810453336_0004/
         user: root
26/03/29 19:40:35 INFO Client: Application report for application_1774810453336_0004 (state: RUNNING)
26/03/29 19:40:35 INFO Client:
         client token: N/A
         diagnostics: N/A
         ApplicationMaster host: cluster-master
         ApplicationMaster RPC port: 37391
         queue: default
         start time: 1774813224861
         final status: UNDEFINED
         tracking URL: http://cluster-master:8088/proxy/application_1774810453336_0004/
         user: root
26/03/29 19:40:51 INFO Client: Application report for application_1774810453336_0004 (state: FINISHED)
26/03/29 19:40:51 INFO Client:
         client token: N/A
         diagnostics: N/A
         ApplicationMaster host: cluster-master
         ApplicationMaster RPC port: 37391
         queue: default
         start time: 1774813224861
         final status: SUCCEEDED
         tracking URL: http://cluster-master:8088/proxy/application_1774810453336_0004/
         user: root
26/03/29 19:40:51 INFO ShutdownHookManager: Shutdown hook called
26/03/29 19:40:51 INFO ShutdownHookManager: Deleting directory /tmp/spark-c87aee1a-a46e-467d-b6da-053394288782
26/03/29 19:40:51 INFO ShutdownHookManager: Deleting directory /tmp/spark-0dae865b-b4c1-40c6-902d-bb2a8b6045ba
3865218 A Briefer History of Time
44671392        A Brief History of Chinese Fiction
60121915        A Brief History of Everyone Who Ever Lived
31318265        A Chance to Make History
1443000 A Brief History of Time (film)
5259052 A Briefer History of Time (Schulman book)
651789  A Calculus of Angels
70599262        A Brief History of Equality
13029312        A Chart of Biography
23727098        A Brief History of Love
```

This screenshot shows the final ranked results for the query `history`.

```
root@cluster-master: /app    root@cluster-master: /app
root@cluster-master:/app# bash add_to_index.sh "/app/New Document.txt"
put: `/app/New Document.txt': No such file or directory
root@cluster-master:/app# printf 'This is a new document for the optional task.\n' > /app/New_Document.txt
root@cluster-master:/app# bash add_to_index.sh /app/New_Document.txt
26/03/29 19:46:47 INFO SparkContext: Running Spark version 3.5.4
26/03/29 19:46:47 INFO SparkContext: OS info Linux, 6.6.87.2-microsoft-standard-WSL2, amd64
26/03/29 19:46:47 INFO SparkContext: Java version 1.8.0_442
26/03/29 19:46:47 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
26/03/29 19:46:47 INFO ResourceUtils: ==============================================================
26/03/29 19:46:47 INFO ResourceUtils: No custom resources configured for spark.driver.
26/03/29 19:46:47 INFO ResourceUtils: ==============================================================
26/03/29 19:46:47 INFO SparkContext: Submitted application: rebuild input
26/03/29 19:46:47 INFO ResourceProfile: Default ResourceProfile created, executor resources: Map(cores -> name: cores, amount: 1, script: , vendor: , memory
 -> name: memory, amount: 1024, script: , vendor: , offHeap -> name: offHeap, amount: 0, script: , vendor: ), task resources: Map(cpus -> name: cpus, amount
: 1.0)
26/03/29 19:46:47 INFO ResourceProfile: Limiting resource is cpu
26/03/29 19:46:47 INFO ResourceProfileManager: Added ResourceProfile id: 0
26/03/29 19:46:47 INFO SecurityManager: Changing view acls to: root
26/03/29 19:46:47 INFO SecurityManager: Changing modify acls to: root
26/03/29 19:46:47 INFO SecurityManager: Changing view acls groups to:
26/03/29 19:46:47 INFO SecurityManager: Changing modify acls groups to:
26/03/29 19:46:47 INFO SecurityManager: SecurityManager: authentication disabled; ui acls disabled; users with view permissions: root; groups with view perm
issions: EMPTY; users with modify permissions: root; groups with modify permissions: EMPTY
26/03/29 19:46:48 INFO Utils: Successfully started service 'sparkDriver' on port 37825.
26/03/29 19:46:48 INFO SparkEnv: Registering MapOutputTracker
26/03/29 19:46:48 INFO SparkEnv: Registering BlockManagerMaster
26/03/29 19:46:48 INFO BlockManagerMasterEndpoint: Using org.apache.spark.storage.DefaultTopologyMapper for getting topology information
26/03/29 19:46:48 INFO BlockManagerMasterEndpoint: BlockManagerMasterEndpoint up
26/03/29 19:46:48 INFO SparkEnv: Registering BlockManagerMasterHeartbeat
26/03/29 19:46:48 INFO DiskBlockManager: Created local directory at /tmp/blockmgr-930bd450-dddb-46ee-95e3-ae7518409eaa
26/03/29 19:46:48 INFO MemoryStore: MemoryStore started with capacity 366.3 MiB
26/03/29 19:46:48 INFO SparkEnv: Registering OutputCommitCoordinator
26/03/29 19:46:48 INFO JettyUtils: Start Jetty 0.0.0.0:4040 for SparkUI
26/03/29 19:46:49 INFO Utils: Successfully started service 'SparkUI' on port 4040.
26/03/29 19:46:49 INFO Executor: Starting executor ID driver on host cluster-master
26/03/29 19:46:49 INFO Executor: OS info Linux, 6.6.87.2-microsoft-standard-WSL2, amd64
26/03/29 19:46:49 INFO Executor: Java version 1.8.0_442
26/03/29 19:46:49 INFO Executor: Starting executor with user classpath (userClassPathFirst = false): ''
26/03/29 19:46:49 INFO Executor: Created or updated repl class loader org.apache.spark.util.MutableURLClassLoader@37946fba for default.
26/03/29 19:46:49 INFO Utils: Successfully started service 'org.apache.spark.network.netty.NettyBlockTransferService' on port 38697.
26/03/29 19:46:49 INFO NettyBlockTransferService: Server created on cluster-master:38697
```

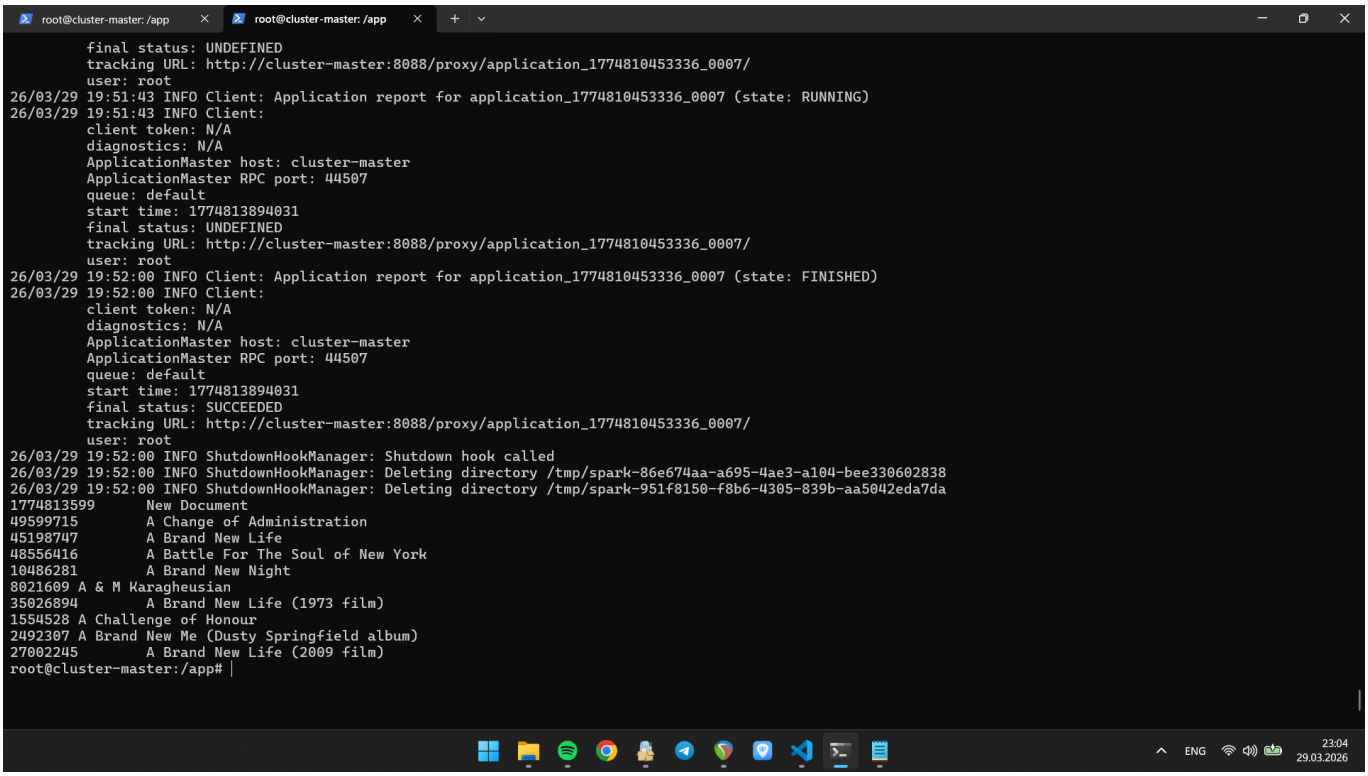
This screenshot shows the optional `add_to_index.sh` task while the new document is being added and the indexing step is running again.

```
root@cluster-master: /app    root@cluster-master: /app
                Input split bytes=222
                Combine input records=0
                Combine output records=0
                Reduce input groups=1
                Reduce shuffle bytes=13057
                Reduce input records=1001
                Reduce output records=1
                Spilled Records=2002
                Shuffled Maps =2
                Failed Shuffles=0
                Merged Map outputs=2
                GC time elapsed (ms)=245
                CPU time spent (ms)=4250
                Physical memory (bytes) snapshot=910893056
                Virtual memory (bytes) snapshot=7773704192
                Total committed heap usage (bytes)=752353280
                Peak Map Physical memory (bytes)=357896192
                Peak Map Virtual memory (bytes)=2590195712
                Peak Reduce Physical memory (bytes)=196014080
                Peak Reduce Virtual memory (bytes)=2593837056
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=16899244
        File Output Format Counters
                Bytes Written=30
2026-03-29 19:47:52,778 INFO streaming.StreamJob: Output directory: /tmp/indexer/pipeline2
Found 4 items
drwxr-xr-x   - root supergroup          0 2026-03-29 19:48 /indexer/documents
drwxr-xr-x   - root supergroup          0 2026-03-29 19:48 /indexer/index
drwxr-xr-x   - root supergroup          0 2026-03-29 19:48 /indexer/stats
drwxr-xr-x   - root supergroup          0 2026-03-29 19:48 /indexer/vocabulary
done storing index
Indexed document:
1774813599      New_Document
root@cluster-master:/app# bash search.sh "new document"
```

This screenshot shows the end of the optional indexing step after the new document was added to the search engine.

```
root@cluster-master: /app    root@cluster-master: /app
1774813599      New_Document
root@cluster-master:/app# bash search.sh "new document"
26/03/29 19:51:31 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
26/03/29 19:51:31 INFO DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at cluster-master/172.18.0.4:8032
26/03/29 19:51:32 INFO Configuration: resource-types.xml not found
26/03/29 19:51:32 INFO ResourceUtils: Unable to find 'resource-types.xml'.
26/03/29 19:51:32 INFO Client: Verifying our application has not requested more than the maximum memory capability of the cluster (8192 MB per container)
26/03/29 19:51:32 INFO Client: Will allocate AM container, with 896 MB memory including 384 MB overhead
26/03/29 19:51:32 INFO Client: Setting up container launch context for our AM
26/03/29 19:51:32 INFO Client: Setting up the launch environment for our AM container
26/03/29 19:51:32 INFO Client: Preparing resources for our AM container
26/03/29 19:51:33 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/HikariCP-2.5.1.jar
26/03/29 19:51:33 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/JLargeArrays-1.5.jar
26/03/29 19:51:33 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/JTransforms-3.1.jar
26/03/29 19:51:33 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/RoaringBitmap-0.9.45
.jar
26/03/29 19:51:33 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/ST4-4.0.4.jar
26/03/29 19:51:33 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/activation-1.1.1.jar
26/03/29 19:51:33 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/aircompressor-0.27.j
ar
26/03/29 19:51:33 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/algebra_2.12-2.0.1.j
ar
26/03/29 19:51:33 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/annotations-17.0.0.j
ar
26/03/29 19:51:33 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/antlr-runtime-3.5.2.
jar
26/03/29 19:51:33 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/antlr4-runtime-4.9.3
.jar
26/03/29 19:51:33 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/aopalliance-repackag
ed-2.6.1.jar
26/03/29 19:51:33 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/arpack-3.0.3.jar
26/03/29 19:51:33 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/arpack_combined_all-
0.1.jar
26/03/29 19:51:33 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/arrow-format-12.0.1.
jar
26/03/29 19:51:33 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/arrow-memory-core-12
.0.1.jar
26/03/29 19:51:33 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/arrow-memory-netty-1
2.0.1.jar
26/03/29 19:51:33 INFO Client: Source and destination file systems are the same. Not copying hdfs://cluster-master:9000/apps/spark/jars/arrow-vector-12.0.1.
jar
```

This screenshot shows the search run for `new document` after the optional task was executed.




This screenshot shows the final search results after adding the new document, which confirms that the optional task updated the index correctly.

This assignment helped me understand how the pieces connect in one full workflow, from preparation, to indexing, to storage, and finally to query-time ranking.